

EMERGING TECHNOLOGIES AND THEIR AREAS

A Comprehensive Report on Emerging Technologies

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Course: DCIT 26 - Applications Development and Emerging Technologies

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EMERGING TECHNOLOGY

- Emerging technologies are the technologies that are new or technologies that are still developing that have the power to create changes in people's lives and society.
Existing technology can also become an emerging technology when it is still improving or used in an innovative way. An example of this is the Artificial Intelligence (AI) that was developed in the 1950s because it is still progressing, and we can use it in different research areas like healthcare, business, education, and other fields.
- According to **Rotolo, Hicks, and Martin (2015)**, emerging technology has 5 main characteristics. It explains how technology emerges.

5 CHARACTERISTICS OF AN EMERGING TECHNOLOGY	
RADICAL NOVELTY	It's a new way of using existing technology. For example, AI is a radical novelty because it can be used in different ways to achieve results.
RELATIVELY FAST GROWTH	Emerging technologies are growing more rapidly than other technologies.
COHERENCE	Technologies that are grouped together to work as one.
PROMINENT IMPACT	These technologies provide benefits across a wide range area like in society.
UNCERTAINTY AND AMBUGUITY	Technologies that are for future uses

HOW EMERGING TECHNOLOGIES HELP AN IT STUDENT

- Emerging Technologies helps computer science and IT students to prepare for a continuously growing field of technology. Learning artificial intelligence, machine learning, IoT, cloud computing, big data analytics, quantum computers, and augmented and virtual reality provides knowledge that can help students that can be used in the industry.
- It demonstrates how technology can be used in different ways. Understanding them helps students to prepare for future development of technology.

EFFECTS OF EMERGING TECHNOLOGY IN THE SOCIETY

- Change the way people live, work, and to interact with technology.
- Makes the life of each other easier and more efficient.
- Artificial Intelligence (AI) can be use in different sectors healthcare, education, work, and many more.

AUGMENTED REALITY (AR) AND VIRTUAL REALITY (VR)

- Augmented Reality (AR) - is the technology that displays a view of the real world into digital with computer-generated sounds, text, and effects.
- Virtual Reality (VR) - a computer generated and 3D environment that a user can explore and interact with that feels real.

ADVANTAGES AND DISADVANTAGES OF ARTIFICIAL INTELLIGENCE

Advantages	Explanation
Reduces Human Effort	AI can do repetitive and time consuming tasks.
Works 24/7	AI can operate continuously.
Technological Development	AI helps technology to grow more advanced.

Can use in different areas	AI can be use in education, work, healthcare, and many more.
Increases Productivity	AI can complete tasks more faster and efficiently.

Disadvantage	Explanation
Job Displacement	AI may reduce opportunities for workers in industries.
Dependence on Technology	AI may reduce human skills and critical thinking.
Limited Creativity	AI follows programmed patterns but lacks imagination.
Security Risk	AI may collect your sensitive data.
Errors	AI can produce inaccurate results.

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Emerging Technologies Report

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RELATIVELY FAST
Improve Rewrite Continue writing Ask AI

Writing assistant
Grammar corrections and writing suggestions

Ensure your writing score is always excellent with unlimited premium recommendations
Upgrade

14 grammar corrections
Accept all

Clarity • Enhance clarity of definition
Premium

Emerging technologies are new technologies or those that are still developing, which have the potential to create significant changes in people's lives and society.
Accept Dismiss

Ensure your writing is always excellent with unlimited Premium recommendations
Upgrade

Fluency • Improve fluency
Premium

Improving or used or is used in

Clarity • Specify range of benefits
Premium

range areas like of areas, particularly in

Grammar 88 Fluency 53 Clarity 55 Engagement 76

Average

History

English

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